

Law and Economics in the Age of Computers

What are the implications of computational complexity for law and economics?

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Yale, Columbia and NBER

Introduction

Law and Economics: Old School and New School Economics

“Not only is it not right, it is not even wrong.” — Wolfgang Pauli (Peierls (1960))

Standard Economic Models

Economics for the Age of Computers - Bounded Rationality and Computational Complexity

A Cognitive Model of Bounded Rationality (Simon (1955), MacLeod (2016))

Bounded Rationality Implies Judicial Discretion (Hart’s Penumbra)

Concluding Remarks

Introduction

- The law and economics movement has helped to organize and improve the law in a wide variety of areas.
- For the most part this has relied on price theory, but there are a number of questions that price theory cannot fully address.
- This lecture breaks down the analytical tools from economics into three distinct classes of models:
 - Price theory (*old school economics*)
 - Information economics (*new school economics*)
 - Bounded Rationality (*contemporary economics and computer science*)

- Each class of models provides a different lens through which to analyze legal and economic issues.
- The choice of model depends on the specific legal and economic issue being analyzed, and whether the framework can lead to better decisions.
- Provide a formal model of bounded rationality consistent with H.L.A. Hart's account of the "penumbra of uncertainty" in legal rules.
Mention values with bounded rationality
- This provides a foundation for understanding how legal rules can be designed to accommodate bounded rationality, ensuring that they are both effective and practically enforceable.

Carry over point from slide 5 of backup: especially for AI

- All complex human societies have developed legal systems to regulate behavior, resolve disputes, and maintain social order.
- These systems have evolved over time to address the changing needs and complexities of human societies, and hence can be viewed, in the sense of Posner (1973), as improving social outcomes over time.

Hammurabi's Code



- Hammurabi's Code is one of the earliest legal codes, dating back to around 1750 BC in ancient Mesopotamia. It is a surprisingly comprehensive set of laws that covers various aspects of society, including family, property, immorality, theft, burglary, wage and price regulation, and even professional conduct.
- See Vincent (1904) for comprehensive discussion of the code.

Implications of Hammurabi's Code for Today

- The code is famously known for its principle of “an eye for an eye,” which emphasizes the idea of retributive justice.
- It also includes provisions for social hierarchy, with different punishments based on the social status of the offender and the victim.
- What is interesting is that many of the main issues have not changed in almost 3,800 years!
- What has changed:
 - New technologies that require new notions of property rights.
 - Better economic theory and evidence to guide the design of better rules (the goal of this lecture).

Law and Economics: Old School and New School Economics

Old School Economics is not so old.

- In contrast to the law, economics is a very new discipline, with what one might consider *modern economics* having its origins in the 1950s with the development of general equilibrium theory, which provides a rigorous mathematical framework for price theory.
- The law and economics movement also started at that time with the founding of the *Journal of Law and Economics* by Aaron Director and Ronald Coase at the University of Chicago.
- The textbook Posner (2014) was very influential in bringing price theory to the law school.
- Other important texts include Baird and Gertner (1994), Cooter and Ulen (2016) and Shavell (2004).

I liked the notes from slide 10 of the backup, where the books are said to combine price theory, game theory, and transactions costs economics for legal rules

New School Economics: Failures of Price Theory due to Imperfect Information i

Year	Laureates	Contribution Summary
2001	George A. Akerlof, A. Michael Spence, Joseph E. Stiglitz	For fundamental analyses of markets with asymmetric information (when buyers or sellers have unequal information).
2007	Leonid Hurwicz, Eric S. Maskin, Roger B. Myerson	For establishing Mechanism Design Theory (how institutions achieve desirable economic outcomes when information is private).
2010	Peter A. Diamond, Dale T. Mortensen, Christopher A. Pissarides	For analyzing markets with search frictions (helping explain why high unemployment can coexist with job vacancies).
2016	Oliver Hart, Bengt Holmström	For foundational contributions to Contract Theory (analyzing executive pay, performance contracts, and privatization).

Year	Laureates	Contribution Summary
2002	Daniel Kahneman, Vernon L. Smith	For integrating psychological research into economic decision-making under uncertainty (Kahneman) and establishing empirical laboratory experiments (Smith).
2017	Richard Thaler	For incorporating psychologically realistic assumptions into economic decision-making (Behavioral Economics).

Contemporary Economics and Computer Science: Analysis of Big Data and Big Problems (Causal and Inductive) i

Year	Laureates	Contribution Summary
2021 (Economics)	David Card, Joshua Angrist, Guido Imbens	For empirical contributions to labor economics (Card) and methodological breakthroughs in analyzing causal relationships using natural experiments (Angrist/Imbens).
2024 (Physics)	John J. Hopfield & Geoffrey E. Hinton	Foundational Neural Network Architectures: Awarded for foundational discoveries that enable machine learning with artificial neural networks. Hopfield created an associative memory (Hopfield Network) that can store and reconstruct patterns. Hinton built on this to invent the Boltzmann machine, paving the way for deep learning architectures that underpin modern generative AI models.

Contemporary Economics and Computer Science: Analysis of Big Data and Big Problems (Causal and Inductive) ii

2024 (Chemistry)	Demis Hassabis & John M. Jumper (shared with David Baker)	AI for Protein Structure Prediction: Awarded for developing AlphaFold2 , an AI system that solved a 50-year-old grand challenge in biology by predicting 3D structures for virtually all known proteins from their amino acid sequences. This revolutionized molecular biology, drug discovery, and biochemical research.
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For the purposes of this lecture, Economics is divided into three groups:

This could probably go before the timeline

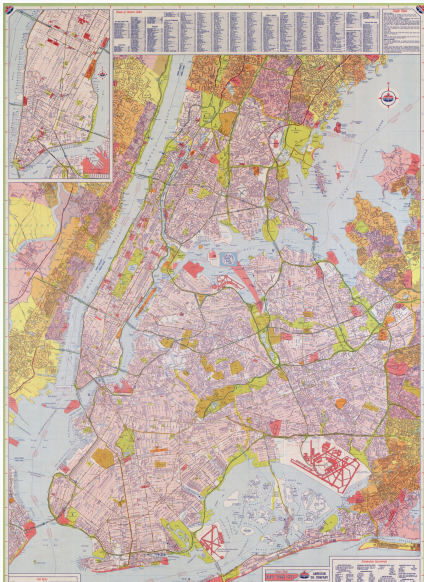
- Price Theory and Game Theory (1940-1980) - the foundations of modern economics (see the appendix of MacLeod (2022)).
- Information Economics, Contract Theory and Causal Analysis (1980-2010)
- Bounded Rationality, Behavioral Economics and Big Data (ideas dating to Simon (1955), but only computationally operational from roughly 2010 on)

“Not only is it not right, it is not even wrong.” — Wolfgang Pauli
(Peierls (1960))

Why do we build models that are clearly wrong?

- One of the most common misunderstandings regarding science is that the goal is to find the “true” model of the world. **This should probably be explained, i.e., no model is correct**
- Models are simplified representations of reality that are *decision-making aids or tools*.
- This is certainly true for physics, where the choice of model depends upon the problem, time and distance scales — from the Planck scales of 10^{-43} seconds (the Planck time) and lengths around 10^{-35} meters (the Planck length) to the cosmological scales where the age of the universe is 10^{10} years and the observable universe has a diameter of around 10^{11} light years.

Example — 1963 Map of New York



- This map provides a lot of information about New York City, including the location of streets, parks, and buildings.
- While you can use it to navigate the subway system, it is “difficult” to use since the print is small, making it hard to find the subway stations.

Not only because the print is small but because subway stations themselves are not laid out in an easily comprehensible manner

Example — Subway Map of New York

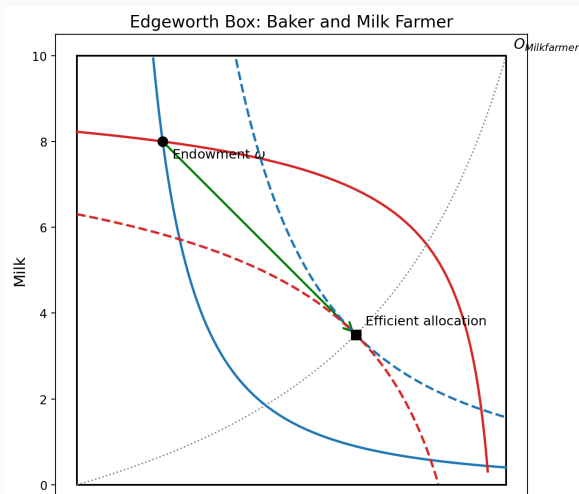


- This is a “wrong” map, but it provides a much more useful guide to choosing a subway route.
- This is a feature of *all* models - they are simplifications of the real world that allow humans to make better decisions.
- A first-order question in many contexts is how we evaluate decision quality.

Standard Economic Models

Price Theory (see appendix of MacLeod (2022))

- When we add prices and require supply = demand, then for any endowment there exists a market allocation (unique in this example).



Price theory is useful, even though it is “wrong”!

- Price theory provides a framework for understanding the allocation of commodities when they are well defined.
- It provides solutions for a number of important policy issues, including pollution (via taxes or tradable permits), congestion pricing (as in London and NYC), and the allocation of property rights via auctions (as in the FCC spectrum auctions, or timber auctions).
- It is the basis for antitrust policy and the control of monopoly power. The extent to which monopoly power is an efficiency or equity issue depends on the nature of information—the topic of today’s lecture.

I would remove the last point about antitrust policy; it might dilute the message

- Model building algorithm:
 - What are the relevant states of Nature (Savage (1972 (first published 1954)))?
 - What are the actions and payoffs available to the decision makers?
 - What are the beliefs of the decision makers over states and actions?
 - Individuals choose (perfect Bayesian / Markov perfect) equilibrium strategies.

e.g., Bayesian/Markov perfect

This theory has been very successful in extending the range of economic analysis.

- Milgrom and Roberts (1992) - Amazing review of how information economics explains organizations.
- Baird and Gertner (1994) - Use information economics to explain many legal rules.
- Gennaioli and Shleifer (2007) - models the evolution of law.
- Listokin (2010) - uses Bayesian methods to set contract damages.
- Chakravarty and MacLeod (2009) - uses information economics to explain the structure of construction contracts.
- Many papers at this conference!

Economics for the Age of Computers - Bounded Rationality and Computational Complexity

Bounded Rationality and Computational Complexity

- Bounded rationality recognizes that individuals have limited cognitive resources and cannot always make fully rational decisions.
- Computational complexity considers the difficulty of solving problems and making decisions given these cognitive limitations. **I start thinking of heuristics and priors: Limited time to compute**
So maybe the “Bounded rationality” and “Computational complexity” points should be flipped
- Together, these concepts help explain why real-world decision-making often deviates from the predictions of standard economic models.

A Cognitive Model of Bounded Rationality (Simon (1955), MacLeod
(2016))

- Chess is a game of perfect information, with a well-defined set of possible states and outcomes.
- Every board position defines the state of the game.
- Given the state, the player whose turn it is chooses a legal move to change the state of the game.
- The winner is the player who checkmates the other player's king.
- There are also rules that ensure that the game always ends, and in some cases a draw is possible.

Chess Playing is an Occupation - what do our previous models have to say about it?

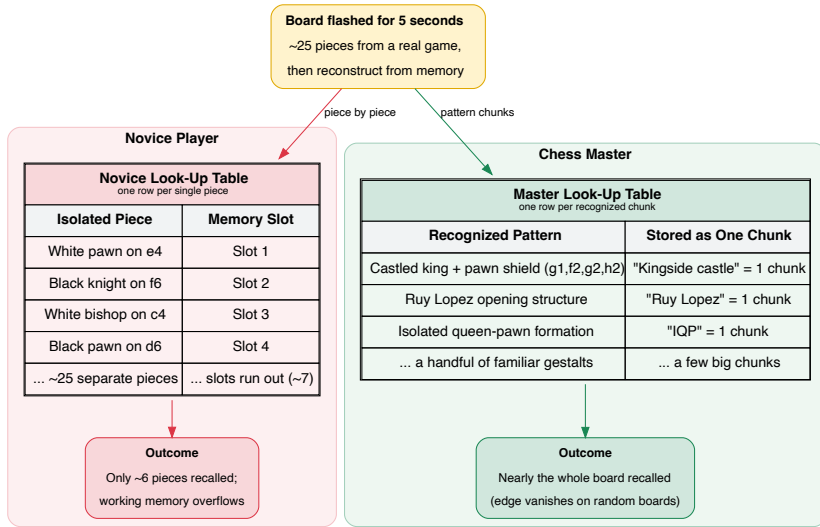
- First, note that there is no uncertainty regarding the rules.
- We can study the game using the tools of information economics - specifically game theory:
 - Zermelo proved that in a game of perfect information like chess, either one player has a winning strategy or both have a drawing strategy.
 - Since the game is finite, and parties observe the state at each stage, the optimal play can be found by backwards induction from the end.
- Yet the game tree has roughly 10^{120} paths (Shannon's estimate), so backward induction is computationally infeasible for any player, human or machine.
Perhaps explicit: Limited time/ability to compute
- As a description of how people actually play, the rational-agent model is clearly rejected by the evidence.

So what do we know about chess skill - Chase and Simon (1973)

Same Position, Two Look-Up Tables:

Novice (square → square) vs. Master (pattern → chunk) [Chase & Simon, 1973]

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.



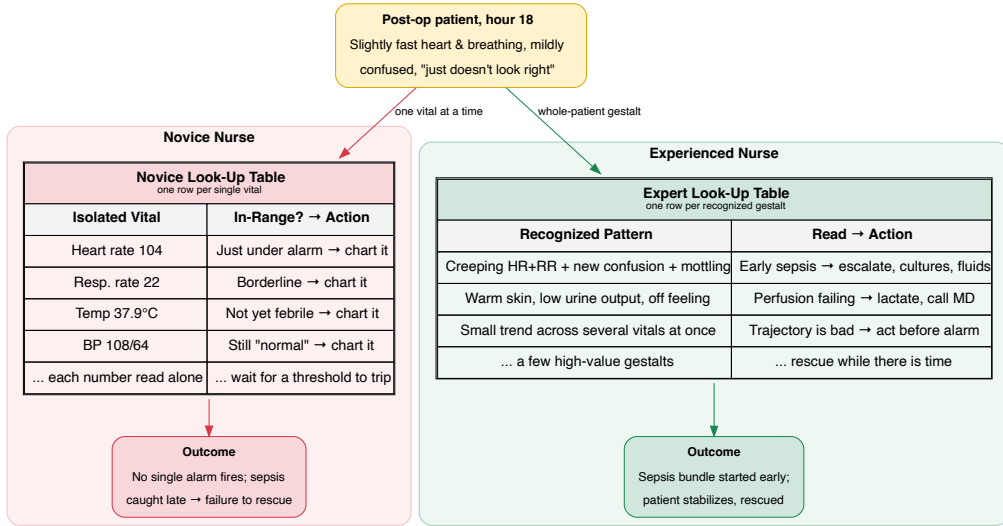
- Chess skill is a form of human capital that is acquired through experience and practice.
- The skill is not a function of the number of possible states, but rather the ability to recognize patterns and to recall the best moves for those patterns.
- This is a form of knowledge that is not easily codified or transmitted, but rather is acquired through learning and practice.
- This has implications for the theory of value: when a player faces a position that is *not* in the lookup table, the choice is guided by an evaluation function (“intuition”) acquired through experience. This *value function* is what fills the gaps in the table.

I think adding back the notes on Churchland and Sejnowski and MacLeod (2016) help to make sense of the following slides and the idea of value development

Same Patient, Two Look-Up Tables:

Novice (vital → normal range) vs. Expert (gestalt → catch sepsis early)

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.

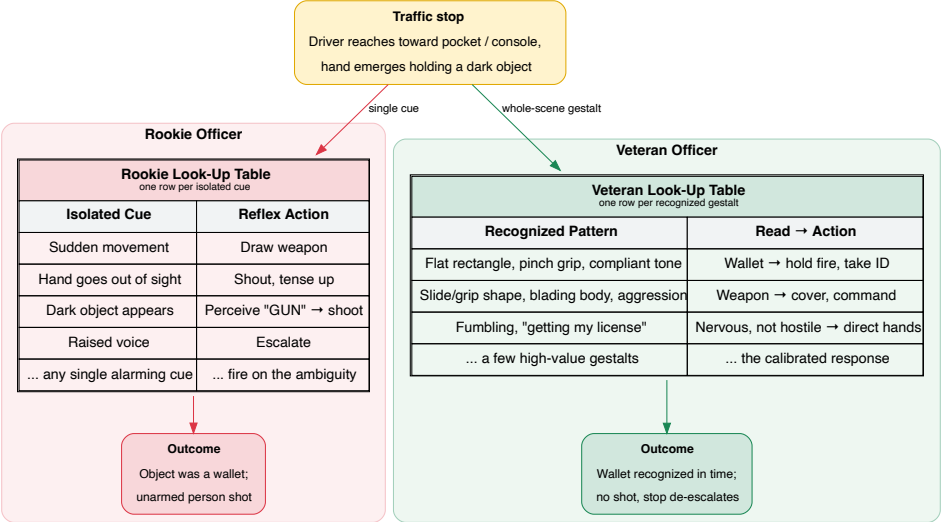


- This example illustrates that nursing skill can affect patient outcomes.
- When an adverse medical event occurs, this may lead to a lawsuit against the hospital.
- The notion of skill is relevant because tort law holds professionals to the standard of care of a reasonably skilled member of their profession (and specialists to a higher standard) when deciding whether there was negligence.

Same Stop, Two Look-Up Tables:

Rookie (cue → shoot) vs. Veteran (gestalt → recognize object)

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.



- This example illustrates that police skill can affect public safety and the outcomes of law enforcement actions.
- When an incident occurs, the skill of the responding officers may determine the resolution and any legal consequences.
- The current U.S. standard for the use of deadly force is “objective reasonableness”: whether a reasonable officer on the scene would have perceived an imminent threat (*Graham v. Connor*, 1989) - not the officer’s subjective belief.

These examples illustrate that both price theory and information economics are not sufficient.

- In all these cases, providing performance rewards at the moment of choice cannot improve outcomes: if the correct action is not in the decision maker's lookup table, no reward can elicit it.
- Rather, optimal choice depends upon rewarding *skill acquisition* ex ante.
- How should we think about aligning AI with human values (“Asilomar AI Principles” (2017))?

Bounded Rationality Implies Judicial Discretion (Hart's Penumbra)

The “Open Texture” of language (Hart (1958))

- Hart (1958) introduced the idea that courts are often faced with cases that fall into the penumbra of uncertainty, where the application of legal rules is not clear-cut.
- This has a number of important implications:
 - Limited time/ability to deliberate
It is impossible for judges to simply “apply the law”.
 - As with the chess player facing a novel position, judges must necessarily attach “human values” to their choices.
 - Like the police officer, judges must make a choice whether or not they are sure what is correct.

The Law as a Lookup Table

- Let $\mathcal{C}$ be the set of possible cases (fact patterns). It is effectively unbounded: new technologies, new relationships and new facts arrive continuously.
- A legal system K (statutes, precedents, doctrines) is a *finite* text that humans can in principle read, with description length $|K|$ bits.
- $Law(K, c) \in \{P, D, \perp\}$ is the outcome K determines for case c : plaintiff, defendant, or *undetermined* ($\perp$).
- $Just(c) \in \{P, D\}$ is the outcome that society's values deem correct - the target that K is trying to encode.
- In the language of the chess example, K is the *lookup table* and $Just$ is the *evaluation function*.

Proposition: Every Finite Legal System has a Penumbra

- **Proposition.** For any finite K , the *penumbra* $\Pi(K) = \{c \in \mathcal{C} : Law(K, c) = \perp \text{ or } Law(K, c) \neq Just(c)\}$ is non-empty.
- **Argument.** A text of $|K|$ bits can distinguish at most $2^{|K|}$ classes of cases. If *Just* varies across more classes than that - as it must when $\mathcal{C}$ is unbounded - then some cases are either undetermined or misclassified. K is a *lossy compression* of *Just*.
- **Corollaries.**
 1. Hart's penumbra is not a defect of drafting; it exists for every finite legal system.
 2. Adding rules shrinks $\Pi(K)$ but never eliminates it, and each new rule creates new boundary cases.
 3. In $\Pi(K)$ the decision-maker must supply a value function - "human values" - to choose between P and D.
 4. An AI system's weights are also a finite K . Hence complete specification of human values is impossible (Hadfield-Menell and Hadfield (2019)); the residual must be handled by *discretion* and by *selecting who exercises it*.

- The proposition is an existence proof for the cases Hart (1958) places in the penumbra of uncertainty.
- For such cases the judge necessarily relies on their *human values* to make a decision.
- Hart illustrates his point with the example of a law that states “No vehicles are allowed in the park.”
- A supervisor who does not like children might use the law to bar women with strollers.
- The women respond by suing the park managers.

The Stroller Example

- A human judge with experience with children and child raising would normally rule in favor of the women with strollers.
- As Gennaioli and Shleifer (2007) discuss, the decision by the human decision maker becomes a legal precedent that makes the law more precise.

How should judges resolve such cases?

- The example illustrates the contrasting values of the park supervisor and the woman.
- In deciding the case the judge must weigh these competing sets of values and interests to arrive at a fair and just decision.
- There are a number of principles that a judge might use - for the moment I will postpone providing a solution.

Rather, let us ask how an AI system might address this problem.

- It has been suggested that AI systems could assist in making legal decisions, as in the case of smart contracts.
- The question here is how would we want an AI system to resolve the stroller case?
- The “Asilomar AI Principles” (“Asilomar AI Principles” (2017)) argue that these decisions should reflect human values. [Instead of , e.g., saying the stroller is a vehicle](#)
- The challenge is that there are a number of principles involved here, including the rule of law, the ordinary meaning of words, and how children and women are affected by the law.

- AI systems are rapidly developing, so one cannot at a given point in time make a statement that will not soon change.
- One can say that recent examples of rogue AI behavior illustrate the challenge of setting goals that result in unintended consequences.
- This is a well-known problem from agency theory - for example the famous study by Kerr (1975).

- Notice that AI firms are attempting to build the “right” incentive model - this is quite similar to the large literature in law and economics on how judges fill gaps in incomplete contracts (e.g. majoritarian defaults, or “what the parties would have agreed upon”).
- What the law actually does is to use an (imperfect) *selection* mechanism:
 - Judges are chosen based on their ability to interpret and apply the law, and on being of “good character”.
 - My work with Elliott Ash (Ash and MacLeod (2015) and Ash and MacLeod (2021)) shows not only that judges have an intrinsic ability to do legal work, but also that non-partisan selection mechanisms can improve the overall quality of the judiciary.

Implications for AI Alignment

- It is increasingly clear that modern AI systems are so complex that even their designers do not really understand their behavior.
- Thus, rather than attempting to specify criteria for all AI systems, we might want to consider how best to *select* systems with desirable properties.
- The key issue is control rights - not over the design of systems, but over the building of selection systems, in much the way we do now for judges.
- The point is quite general - in practice most positions that give individuals control rights are filled via a selection process.

Concluding Remarks

- All socio-economic models are simplifications of the complex environment that is given by modern economies.
- The choice of model depends upon the question being asked.
- In this lecture I have discussed the class of models going back to the work of Herbert Simon in which environmental complexity makes it impossible to anticipate and plan for all contingencies.
- In such environments, decisions necessarily take place in the face of uncertainty.
- This is the world that judges live in.

Summary II - Research Implications

- More work is needed to evaluate judicial selection mechanisms in different contexts and jurisdictions worldwide.
- One is reminded of the great jurists such as Oliver Wendell Holmes, Benjamin Cardozo and Learned Hand, who emphasized the importance of experience and practical wisdom in judicial decision-making.
- Thus, the AI alignment question might be not so much what is the correct algorithm or learning system, but how we build selection processes that choose systems with the experience and practical wisdom that would allow us to delegate to them important life-and-death decisions.

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